WORKED SOLUTION

STEP I 2019, Question 8

Pure • Integration by Substitution

STEP I 2019, Question 8

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

We are told not to evaluate f, so every part is about *reshaping* an integral until it looks exactly like

$$\int_1^X \sqrt{\frac{t-1}{t+1}}\,\mathrm{d}t = \mathrm{f}(X)$$

for some upper limit X. Two moves do almost all the work: a *stretch* ($u = kt$), which turns $\frac{u-k}{u+k}$ into $\frac{t-1}{t+1}$, and a *shift* ($u = t + c$), which makes the two numbers in the fraction symmetric about zero. The last part needs one extra substitution to turn a square root of a quotient into a function of the right shape.

Solution

Throughout, we use the fact that a substitution changes the integrand, the differential *and* the limits; the upper limit is where the answer appears.

Part (i)

The integrand $\sqrt{\frac{u-2}{u+2}}$ becomes $\sqrt{\frac{t-1}{t+1}}$ if we put $u = 2t$, since then $\frac{2t-2}{2t+2} = \frac{t-1}{t+1}$. With $\mathrm{d}u = 2\,\mathrm{d}t$, and limits $u = 2 \mapsto t = 1$, $u = x \mapsto t = \frac{1}{2}x$,

$$\int_2^x \sqrt{\frac{u-2}{u+2}}\,\mathrm{d}u = \int_1^{x/2} \sqrt{\frac{t-1}{t+1}}\,2\,\mathrm{d}t = 2\,\mathrm{f}(\tfrac{1}{2}x).$$

The condition $x > 2$ is exactly what makes $\frac{1}{2}x > 1$, so $\mathrm{f}(\frac{1}{2}x)$ is defined. This is the required result.

Part (ii)

Here the numbers in the fraction are 0 and -4, which are not symmetric about zero. Their midpoint is -2, so we shift by 2: put $u = v - 2$. Then $\frac{u}{u+4} = \frac{v-2}{v+2}$, $\mathrm{d}u = \mathrm{d}v$, and

the limits $u = 0, x$ become $v = 2, x+2$. Now part (i), with $x+2$ (which exceeds 2) in place of x, finishes it:

$$\int_0^x \sqrt{\frac{u}{u+4}}\,\mathrm{d}u = \int_2^{x+2} \sqrt{\frac{v-2}{v+2}}\,\mathrm{d}v = 2\,\mathrm{f}(\tfrac{1}{2}x+1).$$

Part (iii)

The numbers in the fraction are 5 and -1, with midpoint 2. So put $u = v+2$: then

$$\frac{u-5}{u+1} = \frac{v-3}{v+3},$$

and the limits $u = 5, x$ become $v = 3, x-2$. Now stretch by 3, putting $v = 3t$, so that $\frac{v-3}{v+3} = \frac{t-1}{t+1}$ and $\mathrm{d}v = 3\,\mathrm{d}t$, with limits $t = 1, \frac{1}{3}(x-2)$:

$$\begin{aligned}\int_5^x \sqrt{\frac{u-5}{u+1}}\,\mathrm{d}u = \int_3^{x-2} \sqrt{\frac{v-3}{v+3}}\,\mathrm{d}v &= \int_1^{(x-2)/3} \sqrt{\frac{t-1}{t+1}}\,3\,\mathrm{d}t \\ &= 3\,\mathrm{f}\Big(\frac{x-2}{3}\Big).\end{aligned}$$

Since $x > 5$, the argument $\frac{x-2}{3}$ is greater than 1, as it must be. (The two substitutions combine into the single one $u = 3t+2$.)

Part (iv)

This integrand has no square root of a quotient in sight, and the powers are wrong: u^2 on top, u^2+4 under the root.

> **?** **Ask yourself** What substitution would turn u^2+4 into the $u+4$ of part (ii)?

Put $u^2 = w$, so $u = \sqrt{w}$ (we have $u > 0$ on $[1,2]$) and $\mathrm{d}u = \dfrac{\mathrm{d}w}{2\sqrt{w}}$. The limits $u = 1, 2$ become $w = 1, 4$, and

$$\frac{u^2}{\sqrt{u^2+4}}\,\mathrm{d}u = \frac{w}{\sqrt{w+4}} \cdot \frac{\mathrm{d}w}{2\sqrt{w}} = \frac{1}{2} \cdot \frac{\sqrt{w}}{\sqrt{w+4}}\,\mathrm{d}w = \frac{1}{2}\sqrt{\frac{w}{w+4}}\,\mathrm{d}w,$$

which is exactly the integrand of part (ii). Splitting the range at 0 and using part (ii) twice (with $x = 4$ and $x = 1$),

$$\begin{aligned}\int_1^2 \frac{u^2}{\sqrt{u^2+4}}\,\mathrm{d}u &= \frac{1}{2}\left(\int_0^4 \sqrt{\frac{w}{w+4}}\,\mathrm{d}w - \int_0^1 \sqrt{\frac{w}{w+4}}\,\mathrm{d}w\right) \\ &= \frac{1}{2}\Big(2\,\mathrm{f}(3) - 2\,\mathrm{f}(\tfrac{3}{2})\Big) = \mathrm{f}(3) - \mathrm{f}(\tfrac{3}{2}).\end{aligned}$$

Checks

- Each answer has the right behaviour at the lower limit: in (i) at $x = 2$ we get $2\,\mathrm{f}(1) = 0$; in (ii) at $x = 0$ we get $2\,\mathrm{f}(1) = 0$; in (iii) at $x = 5$ we get $3\,\mathrm{f}(1) = 0$.

- Differentiating is a quick test. For (iii), $\frac{\mathrm{d}}{\mathrm{d}x}\,3\,\mathrm{f}\left(\frac{x-2}{3}\right) = \mathrm{f}'\left(\frac{x-2}{3}\right) = \sqrt{\frac{x-5}{x+1}}$, as it should be.
- Numerically, both sides of (iv) equal 0.91007, and in (iii) with $x = 11$ both sides equal 3.19704, each found by direct numerical integration.

Common slips

- Forgetting the factor from $\mathrm{d}u = k\,\mathrm{d}t$, which gives $\mathrm{f}(\frac{1}{2}x)$ instead of $2\,\mathrm{f}(\frac{1}{2}x)$.
- Changing the integrand but not the limits, especially the upper limit, which is where the answer lives.
- In (iv), writing the result as $\frac{1}{2}\cdot 2\,\mathrm{f}(3)$ only: part (ii) gives integrals from 0, so the piece from 0 to 1 must be subtracted.

Beyond STEP

The idea here is that a whole family of integrals is really one integral seen through different affine changes of variable $u = at + b$; recognising the "standard form" hiding inside an expression is a skill used constantly in university analysis and in reducing differential equations to canonical forms. (The function f can in fact be found in closed form using hyperbolic functions, but the question shows that you can get a long way without it.) For the careful theory of substitution and the integral, see Abbott, *Understanding Analysis*; for many more problems in this style, see Siklos, *Advanced Problems in Mathematics*.

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